

Topic 4: Sampling and Aliasing

Summary based on J. H. McClellan, R. W. Schafer, and M. A. Yoder, *Signal Processing First*, Prentice Hall, 2003.

Topics: sampling, normalized frequency, aliasing, the sampling theorem, spectral interpretation, oversampling, stroboscopic aliasing, reconstruction, and interpolation.

Key idea. Sampling converts a continuous-time signal into a discrete sequence, but exact reconstruction is possible only when the sampling rate is high enough to prevent different analogue frequencies from producing the same samples.

4.1 Sampling

A discrete-time signal is represented mathematically as an indexed sequence of numbers. It can be obtained in two ways.

1. By sampling a continuous-time signal $x(t)$. The sequence $x[n]$ is obtained by recording values of $x(t)$ at equally spaced time instants:

$$x[n] = x(nT_s), \quad n \in \mathbb{Z}$$

where T_s is the sampling period and

$$f_s = \frac{1}{T_s}$$

is the sampling rate in samples per second.

2. By calculating the sequence directly from a formula. For example,

$$x[n] = A \cos(\hat{\omega}_0 n + \phi).$$

4.1.1 Sampling sinusoidal signals

For a continuous-time sinusoid

$$x(t) = A \cos(\omega_0 t + \phi),$$

sampling gives

$$x[n] = x(nT_s) = A \cos(\omega_0 nT_s + \phi) = A \cos(\hat{\omega}_0 n + \phi),$$

where the normalized radian frequency is

$$\hat{\omega}_0 = \omega_0 T_s = \frac{\omega_0}{f_s} = 2\pi \frac{f_0}{f_s}.$$

Because ω_0 is measured in radians per second and T_s in seconds, $\hat{\omega}_0$ is measured in radians per sample and is dimensionless.

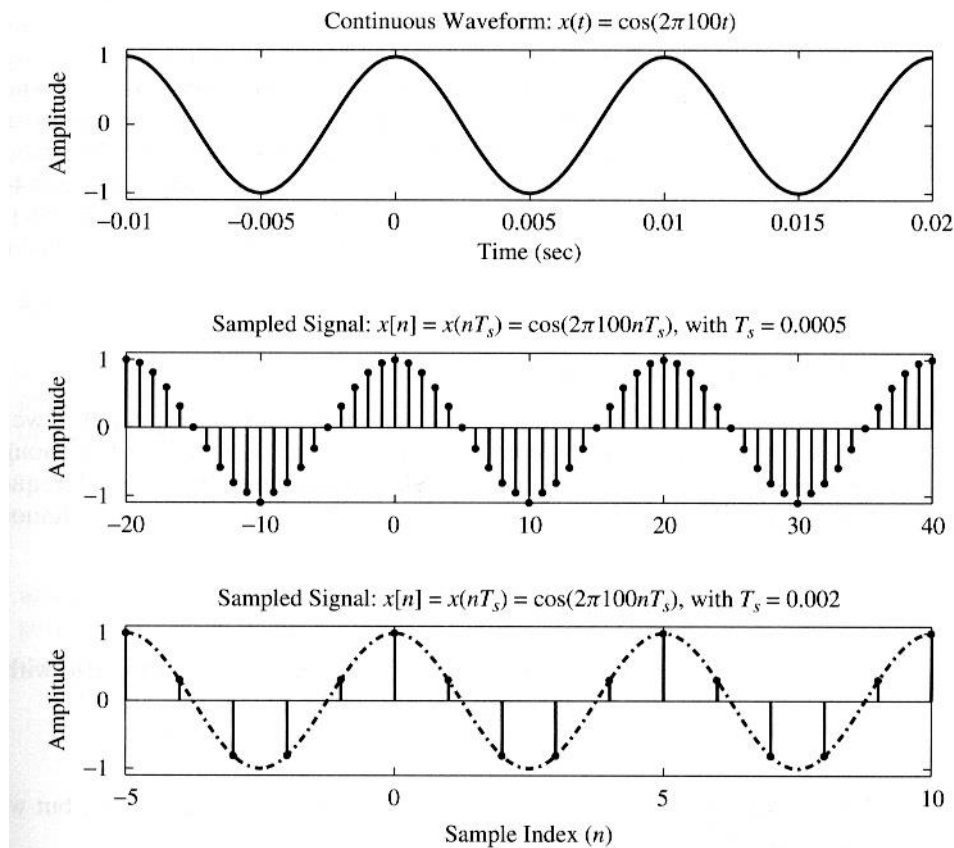


Figure 4.1. A 100-Hz sinusoid sampled at two different sampling rates.

4.1.2 The concept of aliasing

The word *aliasing* comes from the use of two names for the same object. In discrete time, different continuous-time sinusoids can produce exactly the same sample values.

Example. Suppose a sinusoid is sampled at 100 samples/s. Continuous-time frequencies of 20 Hz, 80 Hz, and 120 Hz all give the same sample sequence. From the samples alone, the original analogue frequency cannot be identified without an additional band-limit assumption.

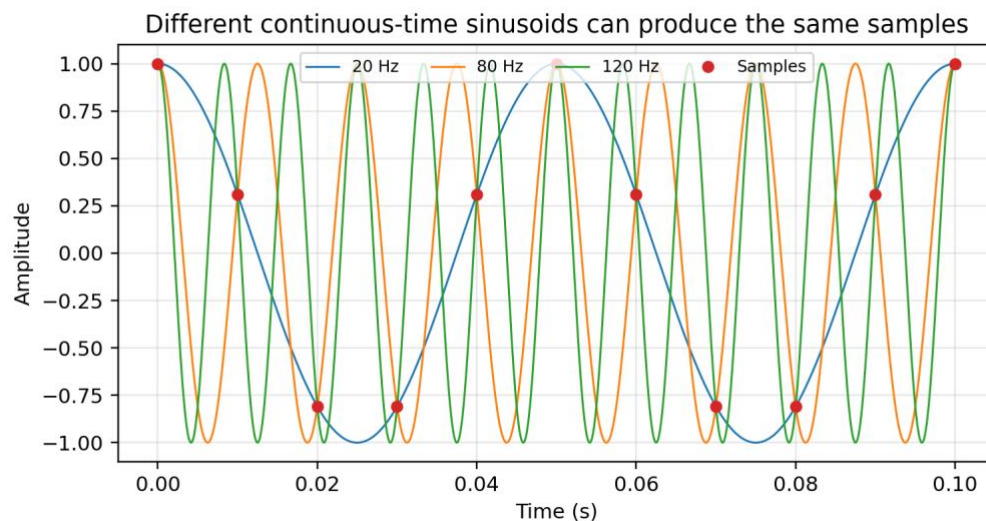


Figure 4.2. Three different continuous-time sinusoids passing through the same sample values.

Since the cosine is 2π -periodic,

$$\cos(\hat{\omega}_0 n + \phi) = \cos((\hat{\omega}_0 + 2\pi k)n + \phi), \quad k \in \mathbb{Z}.$$

Therefore, all frequencies

$$\hat{\omega}_a = \hat{\omega}_0 + 2\pi k$$

are aliases of one another.

Using the identity $\cos(-\theta) = \cos\theta$, another family of aliases is

$$\hat{\omega}_a = -\hat{\omega}_0 + 2\pi k.$$

These are called *folded aliases*. For a general phase,

$$A\cos(-\hat{\omega}_0 n + \phi) = A\cos(\hat{\omega}_0 n - \phi),$$

so folding reverses the sign of the phase.

All aliases can therefore be written compactly as

$$\hat{\omega}_a = \pm\hat{\omega}_0 + 2\pi k, \quad k \in \mathbb{Z}.$$

Equivalently, in hertz,

$$f_a = \pm f_0 + k f_s.$$

4.1.3 The sampling theorem

The key question is how frequently a signal must be sampled so that the original continuous-time signal can be reconstructed from its samples. Shannon's sampling theorem states:

A continuous-time signal $x(t)$ with frequencies no higher than f_{max} can be reconstructed exactly from its samples $x[n] = x(nTs)$ if the samples are taken at a rate $f_s = 1/Ts$ that is greater than $2f_{max}$.

The limiting frequency $f_s/2$ is the Nyquist frequency. Audio CDs use $f_s = 44.1$ kHz, slightly more than twice 20 kHz, the conventional upper limit of human hearing.

4.2 Spectral view of sampling and reconstruction

Sampling changes the apparent locations of spectral components. A sampled sinusoid

Example. For a normalized radian frequency of 0.4π , equivalent spectral lines occur at $0.4\pi + 2\pi k$ and $-0.4\pi + 2\pi k$. Reconstruction selects the representative inside the principal interval from $-\pi$ to π .

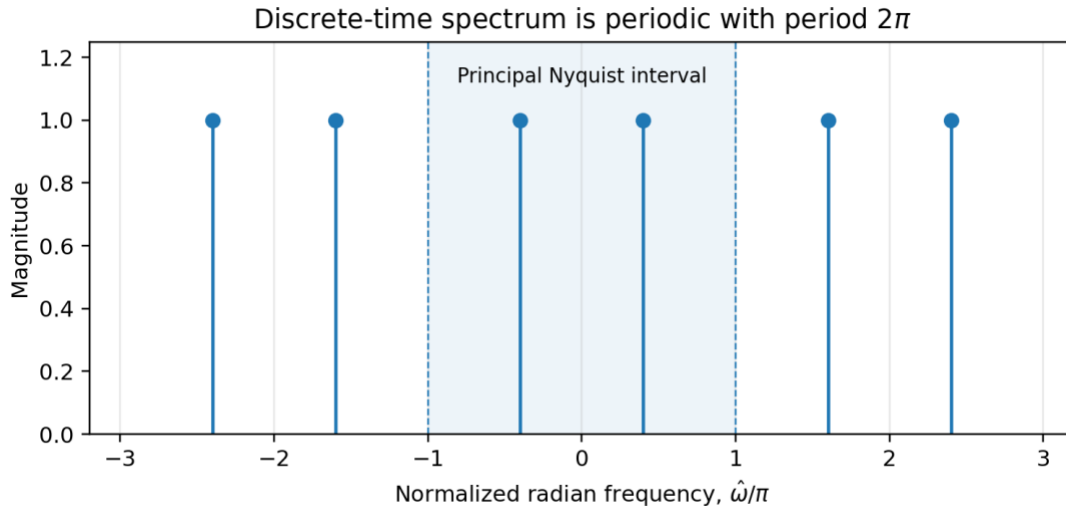


Figure 4.3. Periodic aliases of a discrete-time sinusoid; the shaded region is the principal Nyquist interval.

$$x[n] = A \cos(\hat{\omega}_0 n + \phi)$$

has spectral lines at $\pm \hat{\omega}_0$, but because discrete-time frequency is periodic, it also contains aliases at

$$\hat{\omega} = \pm \hat{\omega}_0 + 2\pi k, \quad k \in \mathbb{Z}.$$

4.2.1 Oversampling

In practice, aliasing is avoided by choosing a sampling rate well above twice the highest signal frequency. This is called *oversampling*.

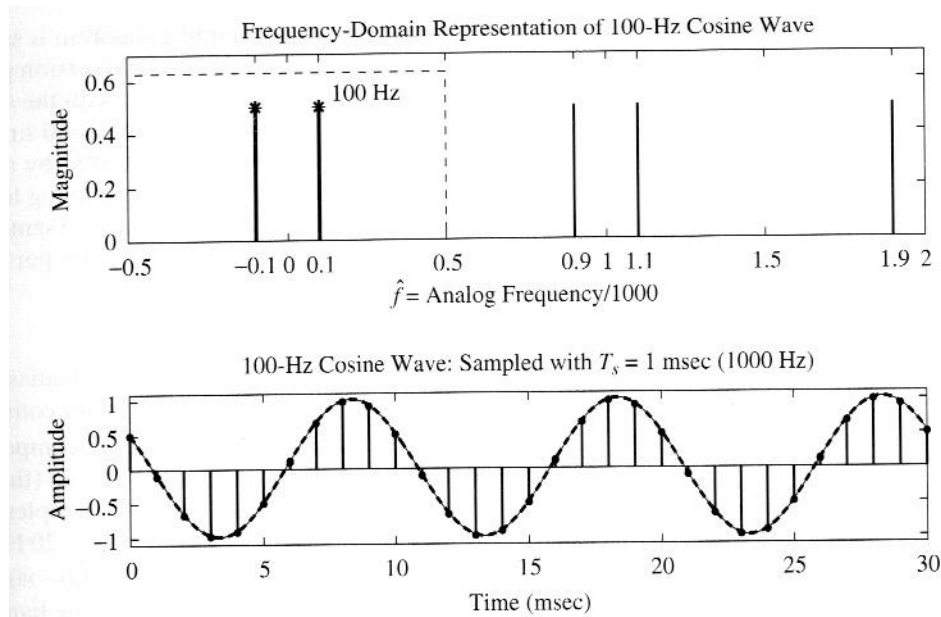


Figure 4.4. Spectral and time-domain views of a 100-Hz sinusoid sampled above the Nyquist rate.

A convenient normalized cyclic frequency is

$$\hat{f} = \frac{f}{f_s},$$

and the corresponding normalized radian frequency is

$$\hat{\omega} = 2\pi\hat{f} = 2\pi\frac{f}{f_s}.$$

Only the frequency interval

$$-\pi < \hat{\omega} \leq \pi$$

or equivalently

$$-\frac{1}{2} < \hat{f} \leq \frac{1}{2}$$

can be reconstructed without ambiguity.

4.2.2 Aliasing caused by undersampling

When $f_s < 2f_0$, a sinusoid is undersampled. A 100-Hz sinusoid sampled at $f_s = 120$ samples/s produces the same samples as a 20-Hz sinusoid because

$$100 = 120 - 20.$$

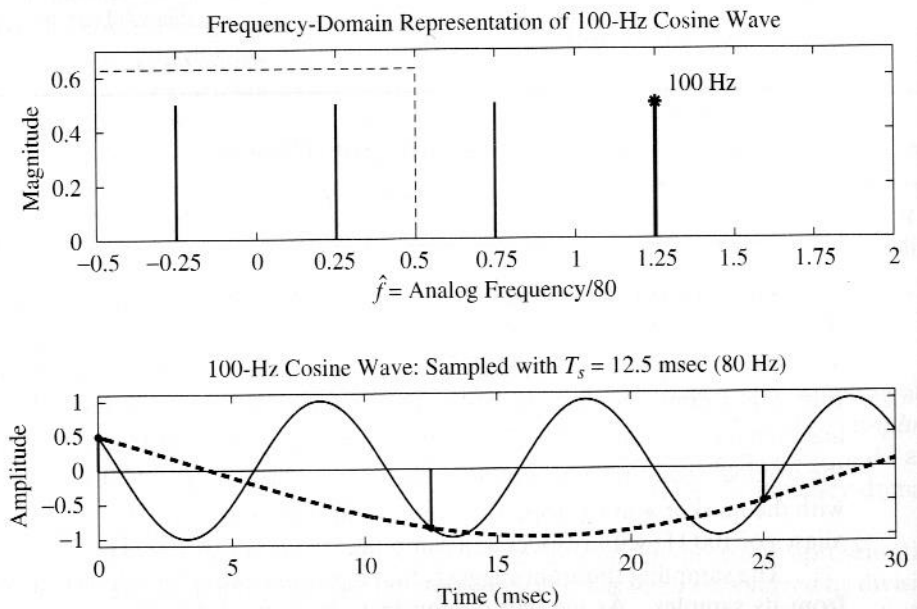


Figure 4.5. A 100-Hz sinusoid sampled at 120 samples/s; the dotted curve is its 20-Hz alias.

If those samples are reconstructed as a low-frequency continuous-time signal, the result is the alias rather than the original sinusoid.

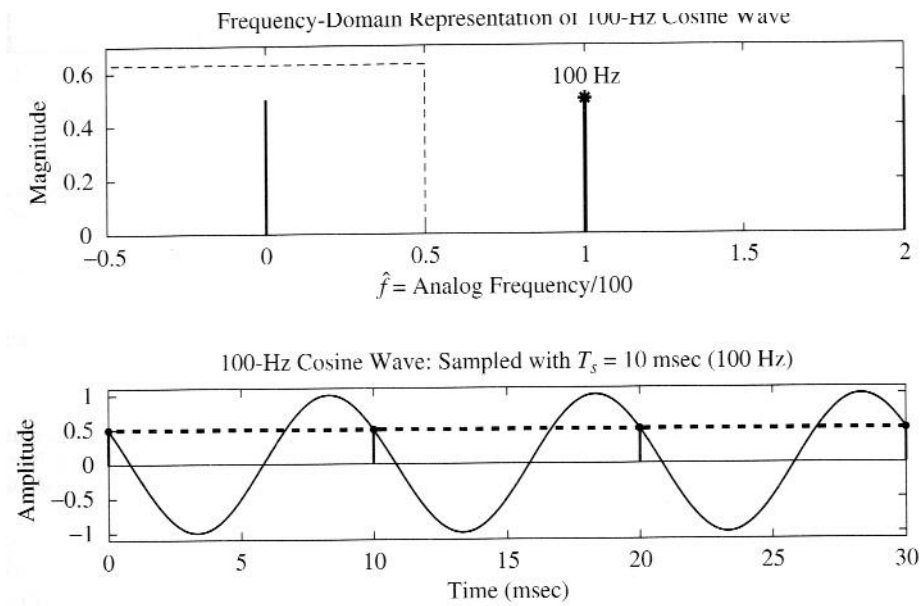


Figure 4.6. Reconstruction of undersampled data produces the lower-frequency alias.

4.2.3 Folding caused by undersampling

At $f_s = 125$ samples/s, a 100-Hz sinusoid folds to 25 Hz:

$$f_a = f_s - f_0 = 125 - 100 = 25 \text{ Hz.}$$

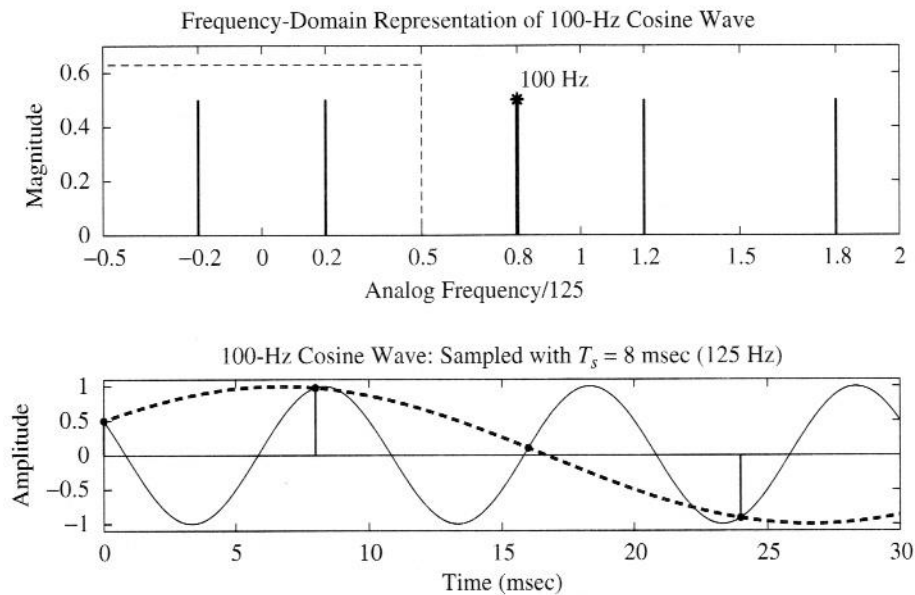


Figure 4.7. A 100-Hz sinusoid sampled at 125 samples/s; the dotted curve is the 25-Hz folded alias.

4.2.4 Maximum reconstruction frequency

The largest uniquely reconstructible frequency is $f_s/2$. Frequencies above this value fold back into the interval $[0, f_s/2]$.

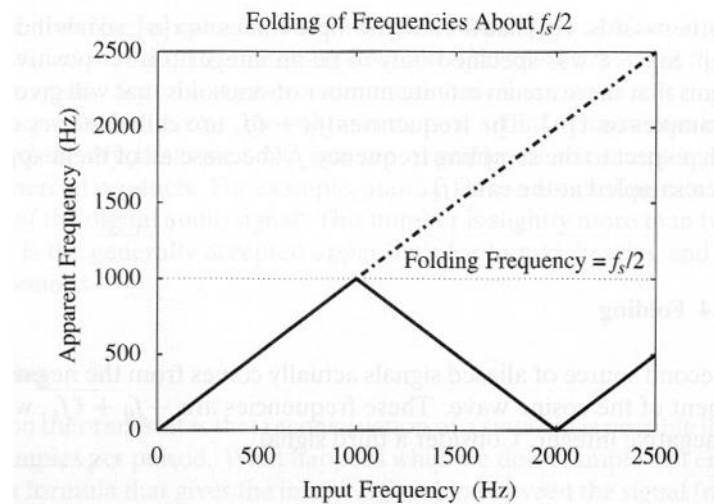


Figure 4.8. Folding of input frequencies about the Nyquist frequency.

4.3 Stroboscopic demonstration

Aliasing can be demonstrated clearly by illuminating a rotating object with a strobe light. Consider a disk attached to a motor shaft and rotating clockwise at constant angular speed.

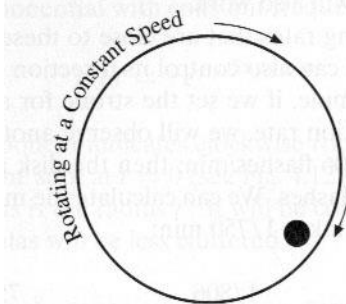


Figure 4.9. Disk rotating clockwise at constant speed.

If the motor speed is 750 rpm and the strobe flashes at $9 \times 750 = 6750$ flashes/min, the black dot advances by

$$\Delta\theta = 2\pi \frac{750}{6750} = \frac{2\pi}{9}$$

per flash.

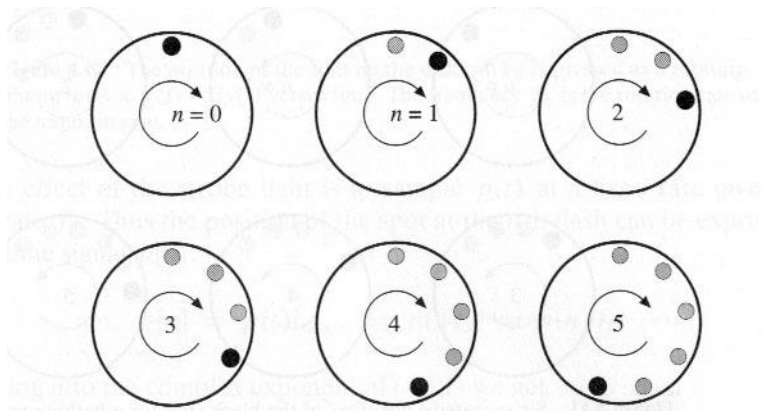


Figure 4.10. Six successive positions of the dot with an angular increment of $\frac{2\pi}{9}$ per flash.

If the flash rate equals the motor speed, the dot appears stationary. The same occurs for flash rates 375, 250, 187.5, and so on. If the flash rate is slightly above one of these values, the dot appears to rotate slowly in the opposite direction.

For example, with 806 flashes/min,

$$\Delta\theta = 2\pi \frac{750}{806} = 2\pi - 2\pi \frac{56}{806},$$

so the observed motion is a small negative rotation at each flash.

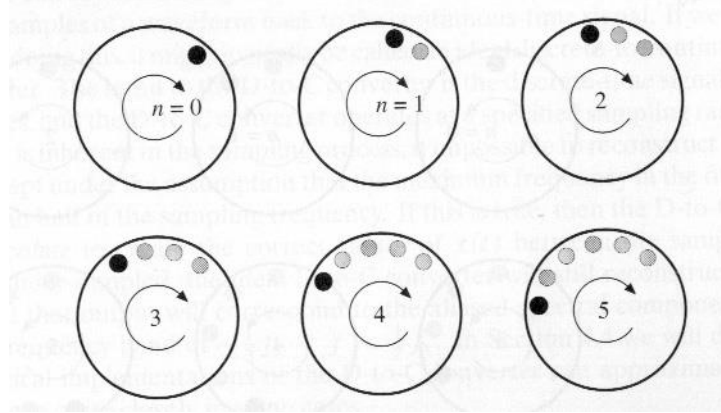


Figure 4.11. Successive positions showing apparent reverse motion caused by aliasing.

The dot position can be represented in Cartesian coordinates as

$$p(t) = x(t) + jy(t) = e^{j\omega_m t},$$

where ω_m is the motor's angular frequency. Sampling with a strobe at period T_s gives

$$p[n] = p(nT_s) = e^{j\omega_m nT_s} = e^{j\hat{\omega}_m n}.$$

If $|\hat{\omega}_m| < \pi$, the sampling theorem is satisfied and there is no aliasing.

4.3.1 Spectral interpretation

A continuously rotating dot has spectral content at the motor frequency.

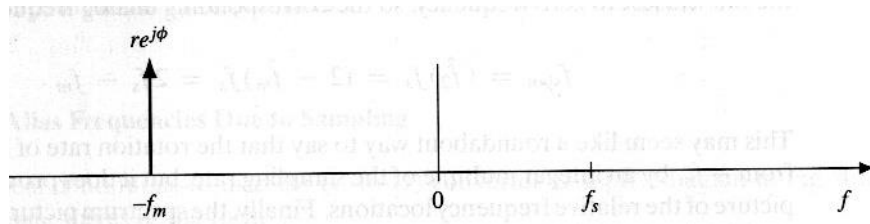


Figure 4.12. Continuous spectrum representation of the rotating disk.

After stroboscopic sampling, the discrete-time spectrum contains an infinite set of aliases:

$$\hat{\omega} = \hat{\omega}_m + 2\pi k, \quad k \in \mathbb{Z}.$$

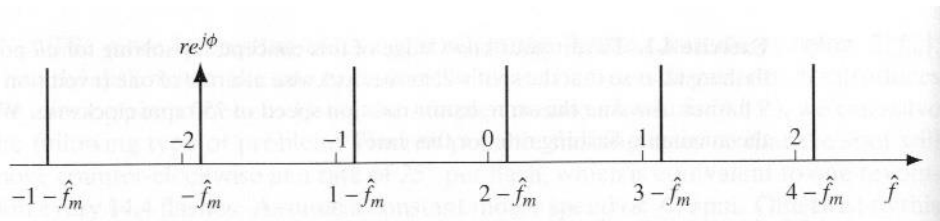


Figure 4.13. Discrete-time spectrum of the rotating disk, showing periodic aliases.

The apparent observed frequency is the alias closest to zero:

$$\hat{\omega}_{\text{obs}} = \hat{\omega}_m - 2\pi \text{round}\left(\frac{\hat{\omega}_m}{2\pi}\right).$$

4.4 Discrete-to-continuous conversion

A digital-to-continuous converter generates a smooth signal from the input samples $x[n]$. If

$$x[n] = A \cos(\hat{\omega}_0 n + \phi)$$

and $|\hat{\omega}_0| < \pi$, ideal reconstruction gives

$$x_r(t) = A \cos\left(\frac{\hat{\omega}_0}{T_s} t + \phi\right).$$

If $|\hat{\omega}_0| > \pi$, the output is distorted by aliasing and the reconstructed sinusoid has the equivalent alias frequency inside the Nyquist interval.

4.4.1 Pulse interpolation

Practical digital-to-analogue converters reconstruct a signal by emitting a pulse for every sample:

$$x_r(t) = \sum_{n=-\infty}^{\infty} x[n] p(t - nT_s),$$

where $p(t)$ is the interpolation pulse. At each time nT_s , a shifted pulse is generated with amplitude proportional to the sample value $x[n]$. The pulse shape strongly affects the reconstructed waveform.

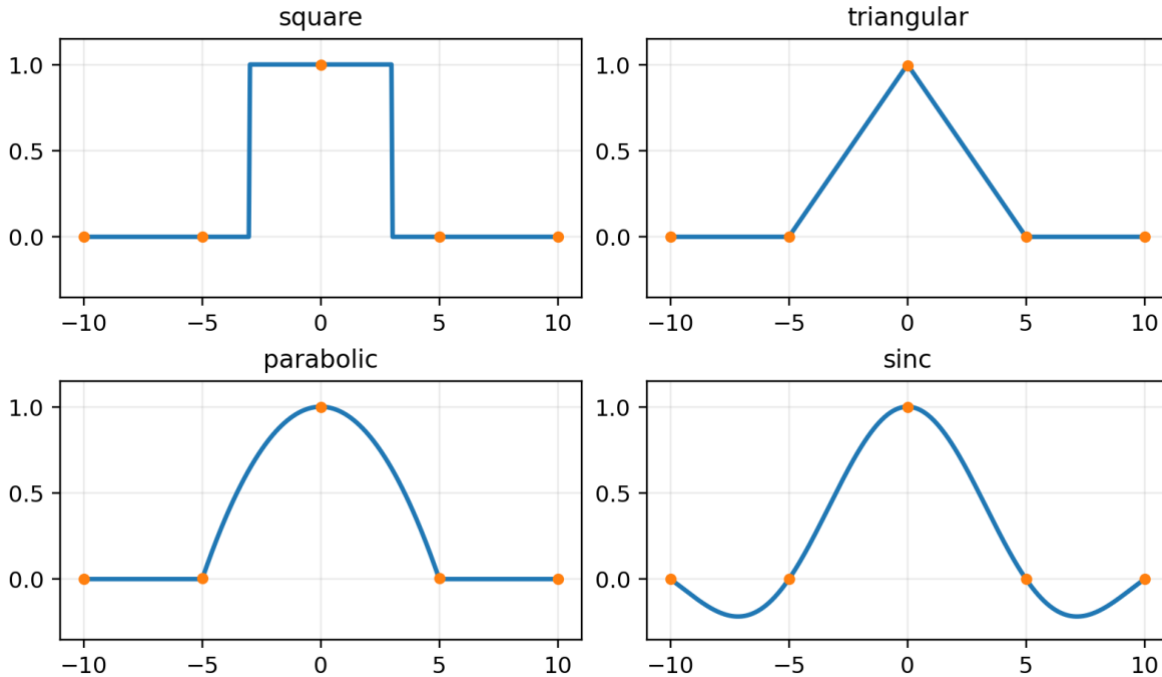


Figure 4.14. Four possible interpolation pulses.

A zero-order hold uses a rectangular pulse and keeps each sample value constant until the next sample.

Example. A sample-and-hold converter produces a staircase waveform. Linear interpolation joins adjacent samples with straight lines and is smoother, although ideal band-limited interpolation is required for exact reconstruction under the sampling theorem.

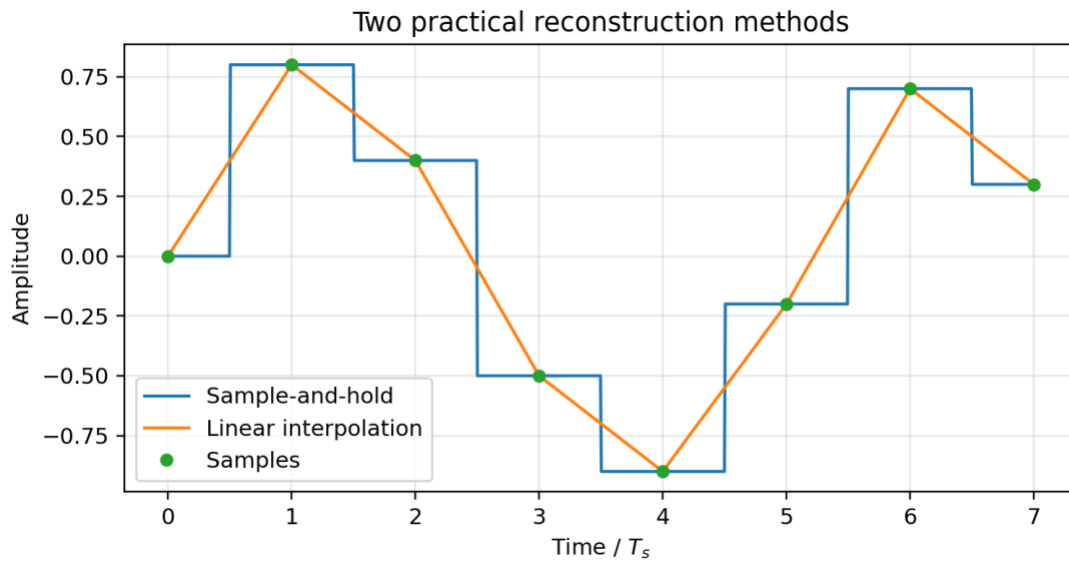


Figure 4.15. Comparison of sample-and-hold and linear interpolation for the same samples.

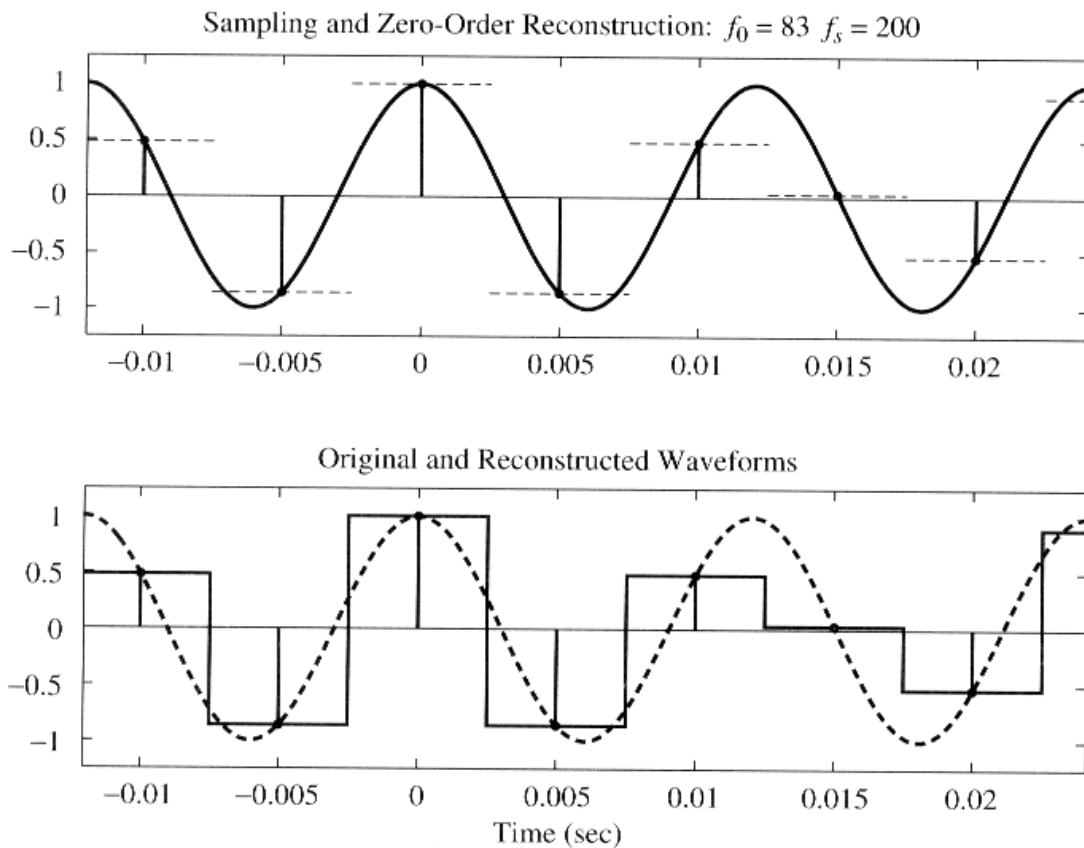


Figure 4.16. Digital-to-continuous conversion using a rectangular pulse.

A smoother pulse, such as a parabolic pulse, gives a smoother interpolation.

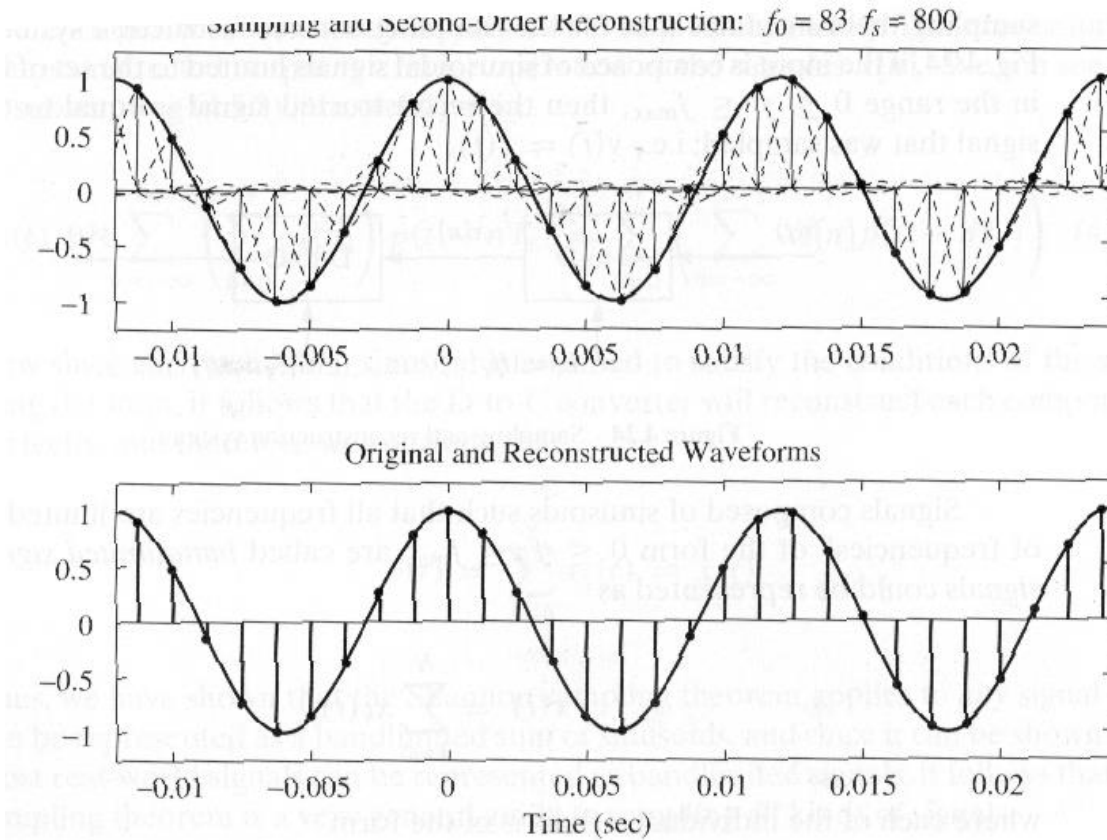


Figure 4.17. Digital-to-continuous conversion using a parabolic pulse.

4.5 Notation

Symbol	Meaning
t	Continuous-time variable (s)
n	Discrete-time sample index
T_s	Sampling period in seconds
$f_s = 1/T_s$	Sampling rate in samples per second (Hz)
f_0	Continuous-time sinusoidal frequency in hertz
$\omega_0 = 2\pi f_0$	Continuous-time radian frequency (rad/s)
$\omega_0 = \omega_0 T_s$	Normalized radian frequency (rad/sample)
$f_0 = f_0 / f_s$	Normalized cyclic frequency (cycles/sample)
$f_N = f_s / 2$	Nyquist frequency
$x(t)$	Continuous-time signal
$x[n] = x(nT_s)$	Sampled discrete-time sequence
$p(t)$	Interpolation pulse used in reconstruction
$\hat{x}(t)$	Reconstructed continuous-time signal